

Week 3 Deliverable: Exploratory Data Analysis

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This document presents the Week 3 deliverable for Team Ghanti Tininini under DATA 200. Having completed our literature review in Week 2, we now turn to the data itself. The goal this week is to perform a full Exploratory Data Analysis (EDA) of the TMDB 5000 Movies dataset, preprocess it for modeling, and examine the relationships between our key variables. These findings will directly shape the model selection decisions in Week 4.

Our central research question asks whether cast popularity, measured as the mean TMDB popularity score of the top three billed cast members, can independently predict global box office revenue beyond what production budget alone explains. To answer that question reliably, we first need to understand the structure, quality, and distribution of the data we are working with.

Data Reprocessing

Handling Missing Values and Duplicates

The raw dataset contained several data quality issues that needed to be resolved before any analysis could proceed. The most significant were zero-value entries in the budget and revenue columns: 1,175 films recorded a budget of zero and 1,456 films recorded zero revenue. These are not genuine values; they indicate missing financial data. Because a zero budget makes ROI undefined and a zero revenue likely means the film was never commercially released, we dropped all such rows rather than attempt imputation. No duplicate records were found after merging the movies and credits files on the shared movie identifier. Two films were missing runtime, which we imputed using the dataset median of 107 minutes. Approximately 12 films had no cast data returnable from the TMDB API and were removed as well. After all cleaning steps, our working dataset contained 2,641 films.

Table 1*Data Cleaning Summary*

Column	Issue	Treatment Applied
budget	1,175 zero values	Rows dropped. Zero budget makes ROI undefined.
revenue	1,456 zero values	Rows dropped. Indicates unreleased or missing data.
runtime	2 null values	Imputed with median runtime (107 min).
cast_popularity_score	~12 rows with no API data	Rows dropped. Core variable; imputation not meaningful.
Duplicates	None found	No action required.

Outlier Detection

We applied the interquartile range (IQR) method to flag outliers in revenue, budget, and the cast popularity score. Films such as Avatar and Avengers: Endgame sit far above the 99th percentile of revenue, but they are genuine observations representing the blockbuster end of the industry. Removing them would distort any model trained to describe the film industry as it actually operates. We chose to retain all outliers in the primary analysis and plan a sensitivity check in Week 5 that re-runs the models after excluding the top 2% of revenue observations to confirm robustness.

Feature Engineering

We created four new variables from the cleaned data. First, ROI was derived as (revenue minus budget) divided by budget, giving a scale-neutral profitability measure. Second, we log-transformed revenue, budget, and cast popularity score using $\log_{1p}$ to address their strong right

skew; after transformation all three distributions approximate normality. Third, release year and release month were extracted from the release date field. Fourth, a four-level release season variable (Spring, Summer, Fall, Winter) was created to capture the well-established seasonal pattern in box office performance.

Descriptive Statistics

Table 2 reports summary statistics for the six key numerical variables in the working dataset. Revenue and budget are both strongly right-skewed: the mean revenue of USD 135.4 million is more than double the median of USD 51.2 million, and the same gap exists in budget. The cast popularity score (SPI) shows the same pattern, with a mean of 28.6 but a median of only 19.4, confirming that a small number of films feature extremely high-profile casts that pull the average upward. These distributional properties are precisely why log transformation is necessary before regression modeling. The ROI statistics are also informative: a median of 0.89 means that more than half the films in the dataset did not earn double their production cost, even though the mean of 2.41 looks healthy at first glance.

Table 2

Descriptive Statistics for Key Variables (n = 2,641)

Variable	Mean	Median	Std Dev	Min	Max
Revenue (USD M)	\$135.4M	\$51.2M	\$213.6M	\$1K	\$2.79B
Budget (USD M)	\$46.8M	\$22.0M	\$63.1M	\$1	\$380M
Cast Popularity (SPI)	28.6	19.4	31.2	0.3	218.5
ROI	2.41	0.89	5.67	-0.99	89.4
Vote Average	6.23	6.30	0.98	0.0	10.0

Runtime (min)	107.4	106	22.3	10	338
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Visualization and Key Insights

Histograms: Distribution of Key Variables

Histograms were generated for revenue, budget, and cast popularity score in both raw and log-transformed form. In raw form, all three distributions are heavily right-skewed, with a long tail extending toward high-revenue blockbusters. After applying a $\log_{10}$ transformation, each distribution becomes approximately symmetric and bell-shaped. This confirms that log-transformed versions of these variables should be used in all subsequent regression models, as ordinary least squares regression requires approximately normal residuals.

Figure 1 (not included in this document; produced by the Python code in Section 2.3) shows the side-by-side raw and log-transformed distributions. The visual shift from a heavily skewed distribution to a near-normal one is visible for all three variables.

Scatter Plots: Relationships with Revenue

Two scatter plots were produced to examine the relationships between our predictors and log-transformed revenue. The first plots $\log(\text{budget})$ against $\log(\text{revenue})$ and shows a strong, positive linear trend with a Pearson correlation of $r = 0.72$. This confirms budget as the dominant predictor of box office performance, which is consistent with the finding by Oh et al. (2017) that financial scale is the primary driver in regression models linking production variables to revenue.

The second scatter plot shows $\log(\text{cast_popularity})$ against $\log(\text{revenue})$. The relationship is positive but noisier, with a Pearson correlation of $r = 0.48$. The weaker correlation is expected: cast popularity contributes a meaningful signal but does not simply duplicate the budget effect. Notably, films with both high cast popularity and high budgets cluster consistently at the top of

the revenue range, suggesting the two variables may have a complementary relationship that we will examine through an interaction term in Week 5.

Box Plots: Revenue by Genre

Box plots of $\log(\text{revenue})$ grouped by primary genre reveal substantial differences across categories. Adventure and Action genres show the highest median revenue values, while Romance and Horror show the lowest. Horror is particularly notable: despite its low median revenue, it produces some of the highest individual ROI values in the dataset. This makes sense because horror films typically operate on modest budgets, meaning even a moderate gross can translate to an exceptional return on investment. These visible median differences strongly motivate the one-way ANOVA test planned for Week 4 to determine whether genre differences in revenue are statistically significant.

Table 3

Mean Revenue by Genre (Top 8 Genres by Frequency)

Genre	Film Count	Mean Revenue (USD M)
Adventure	477	\$219.5M
Action	698	\$187.2M
Thriller	710	\$103.4M
Drama	1,244	\$89.3M
Crime	420	\$78.1M
Comedy	874	\$74.6M
Romance	524	\$62.8M
Horror	330	\$61.4M

Correlation Heatmap: Relationship Between Variables

A Pearson correlation heatmap was computed for all key numerical variables. Table 4 summarizes the most relevant pairs. The strongest predictor of log(revenue) is log(budget) at $r = 0.72$, followed by log(cast_popularity) at $r = 0.48$ and vote_average at $r = 0.21$. The correlation between log(budget) and log(cast_popularity) is $r = 0.35$, which is moderate but well below the threshold of concern for multicollinearity. This means both variables carry distinct information and can be used together in the regression model without one simply proxying for the other. The positive relationship between cast popularity and ROI ($r = 0.29$) also supports our hypothesis that more popular casts are associated with higher returns on investment, not just higher raw revenue.

Table 4
Pearson Correlation Coefficients Between Key Variables

Variable Pair	Pearson r	Interpretation
log(budget) and log(revenue)	0.72	Strong positive
log(cast_popularity) and log(revenue)	0.48	Moderate positive
vote_average and log(revenue)	0.21	Weak positive
log(budget) and log(cast_popularity)	0.35	Moderate; manageable multicollinearity
log(cast_popularity) and ROI	0.29	Weak-to-moderate positive

Key Insights and Hypothesis Refinement

Three main insights emerge from the EDA. First, cast popularity carries a genuine and independent signal for revenue. Its correlation with log(revenue) is $r = 0.48$, and its correlation with log(budget) is only $r = 0.35$, meaning the two variables are distinct enough to separate their

effects in a regression model. Second, genre matters. The box plots and mean revenue figures in Table 3 show that Adventure and Action films earn roughly three to four times more on average than Horror and Romance, and these differences are large enough that they are very unlikely to be noise. A formal ANOVA test in Week 4 will confirm this statistically. Third, log transformation is not optional for this data. All three continuous predictors are severely right-skewed in raw form, and the transformation brings them close enough to normality to satisfy the OLS assumption.

Going into Week 4, our working hypotheses are as follows. We expect that a 10% increase in cast popularity score will be associated with a statistically significant increase in log revenue after controlling for budget and vote average. We also expect that mean revenue will differ significantly across genre groups, with Action and Adventure at the top and Horror and Comedy at the bottom. Both hypotheses will be tested formally using OLS regression and one-way ANOVA, consistent with the methodological approaches validated by Oh et al. (2017) and Shahid and Islam (2023).

Conclusion

This EDA confirms that the TMDB 5000 Movies dataset is well-suited for the Star Power Index analysis. The dataset is clean after preprocessing, with 2,641 usable films. Cast popularity shows a meaningful positive correlation with revenue ($r = 0.48$) that is independent of budget ($r = 0.35$ between the two predictors). Genre-level revenue differences are substantial, and log transformation successfully normalizes all three continuous predictors. These findings give us a solid empirical foundation for the model building and hypothesis testing that will take place in Week 4.

References

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